

Depth, Breadth, and Valuation: A Unified Ontology of the Physical Continuum

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Abstract

The standard mathematical ontology of the continuum — the real numbers with their uncountable cardinality and Dedekind-complete order — conflates three distinct properties with different physical status. We decompose the continuum into three axes: depth (Archimedean completeness), breadth (set-theoretic uncountability), and valuation (ultrametric completions of the rational numbers). Depth is physically real — it is required for dynamics, causality, and connectedness. Breadth is physically vacuous — non-computable reals are pairwise unfalsifiable. Valuation is physically real but not geometric — p-adic completions encode discrete quantum structure (spin, internal numbers, information) with an operational ontology distinct from the Archimedean continuum. The complete physical continuum is the real numbers $\times \prod_p \mathbb{Q}_p$ — the rational numbers $\hat{\mathbb{Q}}$ — the computable reals together with the computable p-adic numbers at finitely many primes. Ostrowski's theorem is the deep organizing principle: there are no other completions of the rational numbers, so there are no other physically real dimensions of the continuum. The resulting framework — adelic ontological closure — makes specific, falsifiable predictions that no Archimedean-only theory can make.

1. Introduction: The 2D Continuum Model Decomposed

1.1 The Original Insight

A heuristic model decomposes “uncountability” into two dimensions:

- **Depth:** Infinite fineness — between any two numbers, there is another. This is the Archimedean property, the connectedness of the continuum, the ability to take limits.
- **Breadth:** Cardinality explosion — Hilbert's Hotel, Cantor's diagonal, the vast majority of real numbers that can never be individually named.

The model's key insight: these are distinct properties, often conflated. The question it raises: do depth and breadth exist in physical reality, or are they mathematical artifacts?

Papers I and II answered this question in detail. Paper III synthesizes the answers into a unified ontological framework, decomposes the 2D model into a complete 3-axis picture, and establishes the operational criterion by which each axis is accepted or rejected.

1.2 The Three-Axis Framework

The completions of $\mathbb{Q}$, by Ostrowski's theorem, partition into:

Axis	Mathematical Object	Physical Status	Key Property
Depth	Archimedean completion $\mathbb{R}$, computable subset $\mathbb{R}_c$	Physically real	Connectedness, dynamics, causality, limits
Breadth	$\mathbb{R} \setminus \mathbb{R}_c$ — the non-computable reals	Physically vacuous	Unfalsifiable, no measurement- distinguishable predictions
Valuation	$\mathbb{Q}_p$ for primes p , computable subset $\mathbb{Q}_p^c$	Physically real (as information carriers)	Discrete quantum numbers, spin, hierarchical information structure

This replaces the 2D “depth vs. breadth” model with a 3-axis framework where **breadth is eliminated entirely** and **valuation is recognized as a distinct physical category** — neither depth (it is not Archimedean/connected) nor breadth (it makes falsifiable predictions).

2. Depth: The Archimedean Completion $\mathbb{R}_c$

2.1 What It Is

The Archimedean completion of $\mathbb{Q}$ is the unique connected, ordered, complete (in the Cauchy sense) field containing $\mathbb{Q}$. Its computable subset $\mathbb{R}_c$ preserves all physically relevant properties:

Property	Holds in $\mathbb{R}_c$?	Physical Role
Ordered field	PASS	Basic arithmetic of physical quantities
Computable completeness	PASS	All measurement sequences with computable error bounds converge
Connectedness	PASS	Causality, topology, intermediate value theorem
Real closure	PASS	Algebraic closure of physically real numbers
Uncountability	FAIL	Not required —

2.2 What It Does

Depth is the continuum of **time and dynamics**. It is the substrate on which fields evolve, particles propagate, and causality operates. Its key features — connectedness and computable completeness — are necessary for physics:

- Without connectedness, there is no intermediate value theorem, no guarantee that a continuous quantity passes through all intermediate values, no basis for classical dynamics.
- Without computable completeness, limits of measurement sequences need not exist, and the notion of “arbitrarily precise measurement” is incoherent.

Depth is the minimal completion of $\mathbb{Q}$ required for dynamics. And $\mathbb{R}_c$ — not $\mathbb{R}$ — is that completion.

2.3 What It Does NOT Require

Depth does not require:

- Uncountability ($|\mathbb{R}_c| = \aleph_0$)
 - The Continuum Hypothesis (distinction between $\aleph_1$ and $\aleph_2$ is physically inert)
 - Non-computable reals (they are pairwise physically indistinguishable — Paper I, Theorem 4.3)
 - Dedekind completeness (computable Cauchy completeness suffices for all measurement sequences)
 - Chaitin's Ω (its binary expansion can never be measured; it is a proof-theoretic artifact)
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3. Breadth: The Dark Continuum $\mathbb{R} \setminus \mathbb{R}_c$

3.1 What It Is

The complement $\mathbb{R} \setminus \mathbb{R}_c$ consists of all non-computable real numbers — reals for which there is no Turing machine that approximates them with a computable modulus of convergence. This set has cardinality $2^{\aleph_0}$ (continuum).

3.2 Why It Is Physically Vacuous

Paper I, Theorem 4.3 (Unfalsifiability of Non-Computable Reals) establishes:

For any two non-computable reals x, y that agree on all rational approximations with computable error bounds, no finite measurement protocol can produce distinct predictions parameterized by x versus y .

The proof is constructive: any measurement protocol defines a computable functional, and computable functionals cannot distinguish between reals that agree on computable tests. The excess reals in $\mathbb{R} \setminus \mathbb{R}_c$ are physical duplicates of their computable shadows — the nearest computable rational approximations.

The dark continuum is physically empty. It adds cardinality to the mathematical continuum but no information to the physical one.

3.3 What This Means for Mathematics

This is not a rejection of classical mathematics. ZFC proves true things about $\mathbb{R}$ that are physically vacuous. The distinction between $\mathbb{R}$ and $\mathbb{R}_c$ is the distinction between a mathematical universe that includes the full power set of $\mathbb{N}$ and one that includes only the computable subsets. The latter is sufficient for physics; the former is richer but physically inert.

The Continuum Hypothesis, the Banach-Tarski paradox, Chaitin's Ω — these are true (or independent) in ZFC but have zero physical content. They quantify over objects — non-computable subsets, non-measurable partitions, non-computable reals — that are not physically accessible.

4. Valuation: The Ultrametric Completions $\mathbb{Q}_p$

4.1 What They Are

For each prime p , the p -adic numbers $\mathbb{Q}_p$ form a complete, ultrametric field containing $\mathbb{Q}$. Unlike $\mathbb{R}$, they are totally disconnected — their topology is a Cantor set. Their geometry is tree-structured: the Bruhat-Tits building for $SL_n(\mathbb{Q}_p)$ is a simplicial complex encoding the lattice-theoretic structure of the completion.

The computable subset $\mathbb{Q}_p^c$ (p -adic numbers approximable by Turing machines) is countable and sufficient for physical content — paralleling the $\mathbb{R}_c/\mathbb{R}$ distinction.

4.2 Physical Role: Information, Not Geometry

The p -adic completions are physically real, but their role is **not geometric** — they are not “extra spatial dimensions.” They are **valuation-theoretic dimensions** encoding discrete, hierarchical, information-theoretic structure:

p	Physical Role	Realization
2	Spin, fermion/boson statistics	$\mathbb{Z}_2$ grading on Bruhat-Tits tree boundary
3	Color charge (QCD)	Vertex labels in $SL_3(\mathbb{Q}_3)$ Bruhat-Tits building
Other p	Additional discrete quantum numbers	Bruhat-Tits buildings at the corresponding primes

The key insight from Paper II, §5.1:

The Archimedean completion gives you time, continuity, fields, and causal structure. The p -adic completions give you spin, charge, color, and entanglement — the discrete, hierarchical, information-theoretic properties that resist Archimedean description.

4.3 Why Valuation Is NOT Breadth

A natural objection: each $\mathbb{Q}_p$ has cardinality $2^{\aleph_0}$ (continuum), just like $\mathbb{R}$. Doesn’t the breadth problem apply to p -adic numbers too?

No. The distinction is operational:

- Non-computable reals ($\mathbb{R} \setminus \mathbb{R}_c$) are physically indistinguishable from each other and from their computable shadows. They add no measurement-distinguishable content (Paper I, Theorem 4.3).
- Non-computable p -adic numbers ($\mathbb{Q}_p \setminus \mathbb{Q}_p^c$) face the same objection — they too are physically vacuous. The computable p -adic numbers $\mathbb{Q}_p^c$ are sufficient for physical content.
- **But the computable p -adic numbers make falsifiable physical predictions** — Gromov $\delta = 0$ for ZBW transitions, $\mathbb{Z}_2$ invariant distinguishing Dirac from Majorana, p -adic valuation gaps between optimal and random codes. These predictions are testable independent of any non-computable structure.

Valuation is physically real (through $\mathbb{Q}_p^c$) because it generates measurement-distinguishable predictions. Breadth is vacuous regardless of which completion it appears in.

5. The Complete Ontological Criterion

5.1 The OC Criterion, Unified

Definition 5.1 (Physically Real Entity). An entity is physically real iff there exists a finite D/R (description/realization) procedure — a Turing machine — that: 1. Approximates the entity with a computable modulus of convergence, and 2. Generates measurement-distinguishable predictions that differ from the predictions of at least one alternative (i.e., the entity is falsifiable).

Clause (1) eliminates the dark continuum. Clause (2) eliminates entities that, while computable, make no empirical difference.

5.2 Application to Each Axis

Axis	Satisfies (1)?	Satisfies (2)?	Physical Status
Depth ($\mathbb{R}_c$)	PASS — all $x \in \mathbb{R}_c$ are computable by definition	PASS — different computable reals produce different measurement sequences	Physically real
Breadth ($\mathbb{R} \setminus \mathbb{R}_c$)	FAIL — no finite procedure approximates them	FAIL — they are unfalsifiable (Theorem 4.3, Paper I)	Physically vacuous
Valuation ($\mathbb{Q}_p^c$)	PASS — computable p-adic numbers are Turing-approximable	PASS — falsifiable predictions (Gromov δ , $\mathbb{Z}_2$ invariant, valuation gaps)	Physically real
Valuation breadth ($\mathbb{Q}_p \setminus \mathbb{Q}_p^c$)	FAIL — same as breadth above	FAIL — same as breadth above	Physically vacuous

5.3 The Complete Physical Continuum

The operational ontology of the physical continuum is:

$$\mathcal{C}_{\text{phys}} = \mathbb{R}_c \times \prod_{p \in S} \mathbb{Q}_p^c$$

where S is the (finite) set of primes whose p -adic completions encode physically relevant quantum structure.

At minimum, $S \supseteq \{2, 3\}$ (spin and color). Additional primes may encode additional discrete quantum numbers — the empirical question is open.

$\mathbb{A}_{\mathbb{Q}}^c$ — the ring of computable adeles — is the candidate framework for a unified adelic physical theory: a product of the computable reals and all computable p -adic numbers, restricted to finitely many non-Archimedean places for any given physical system.

6. Beyond the 2D Model: Falsifiable Predictions

6.1 Predictions from the Archimedean Reduction (Paper I)

1. **No measurement can distinguish a non-computable real from its computable shadow.** Any experiment claiming to measure a non-computable quantity (e.g., a value that depends on the halting problem) is measuring an illusion — the quantity actually measured is computable.
2. **The Continuum Hypothesis cannot affect any measurement outcome.** Any theory whose predictions change depending on whether $|\mathbb{R}| = \aleph_1$ or $|\mathbb{R}| > \aleph_1$ is not a physical theory — it is a mathematical artifact.

6.2 Predictions from the P-adic Framework (Paper II)

1. **Gromov $\delta = 0$ for ZBW transitions.** The Zitterbewegung current correlator is a tree (not merely tree-like). Spin noise spectroscopy or EELS/RIXS can measure this.
2. **$\mathbb{Z}_2$ invariant distinguishes Dirac (invariant = +1) from Majorana (invariant = -1).** This is a momentum-dependent diagnostic at the field-theoretic level — testable in condensed matter systems hosting Majorana zero modes.
3. **Optimal quantum error-correcting codes exhibit p -adic valuation gaps.** $v_p^{\{\max\}}$ is significantly higher for optimal codes than random ones ($7\times$ for known families). This is testable by constructing code families and measuring their valuations.

6.3 Predictions from the Unified Framework (this paper)

1. **No Archimedean perturbation can affect p -adic fixed points.** The mutual singularity of $\mathbb{R}$ and $\mathbb{Q}_p$ topologies (Ostrowski) implies that Majorana zero modes on Bruhat-Tits trees are immune to thermal, electromagnetic, and vibrational decoherence — all of which are Archimedean perturbations.
2. **Physics is finite-dimensional over $\mathbb{Q}$ at finitely many places.** The number of physically relevant primes S is finite. This constrains the possible structure of beyond-Standard-Model theories: they may add new primes but not a continuous infinity of them.

3. **The JPCUB (Joules-per-CUBit) metric applies.** Computational advantage is measured not in abstract gate counts but in joules per solution at commercially relevant scale. Systems exploiting p-adic intrinsic protection (adelic QEC) should beat surface-code-based quantum computers on JPCUB by avoiding the 10^2 – 10^3 error-correction overhead.
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7. Implications

7.1 For Mathematics

The 3-axis framework provides a principled demarcation between mathematical structures with physical content and those without:

- **Physical:** computable reals ($\mathbb{R}_c$), computable p-adic numbers at finitely many primes ($\mathbb{Q}_p^c$), and their finite products.
- **Mathematical but physically vacuous:** the full power set of $\mathbb{N}$, non-computable reals, the Continuum Hypothesis, and other set-theoretic structures that quantify over physically inaccessible objects.

This is not a limitation on mathematics. It is a limitation on what mathematics can contribute to physics without specifying an operational protocol for measurement.

7.2 For Physics

The unified framework suggests:

1. **Quantum field theory should be reformulated over the ring of computable adèles $A_{\mathbb{Q}}^c$, not over $\mathbb{R}^4$.** The familiar QFT over $\mathbb{R}^{3,1}$ is the Archimedean projection of a richer adelic theory.
2. **The Standard Model’s gauge group is not arbitrary.** $SU(3)_C \times SU(2)_L \times U(1)_Y$ decomposes by prime factors $(3, 2, \infty)$ because the completions at those places are the physically relevant ones. New gauge symmetries, if any, should appear at additional primes.
3. **The measurement problem may have a number-theoretic resolution.** Wavefunction collapse from a superposition to a definite outcome corresponds to a transition from the Archimedean description (continuous amplitudes) to the p-adic description (discrete measurement outcomes). This is speculative but falsifiable: if measurement outcomes are p-adic valuations, the distribution of outcomes should follow p-adic statistics, not Archimedean ones.

7.3 For Philosophy of Science

The 3-axis framework operationalizes the distinction between “exists mathematically” and “exists physically”:

- **Mathematical existence** = consistency with ZFC (or equivalent foundation).
- **Physical existence** = finite Turing-machine approximability with computable modulus + generation of measurement-distinguishable predictions.

The dark continuum exists mathematically but not physically. The computable p-adic numbers exist both mathematically and physically. The computable reals exist physically and are sufficient

for all dynamical predictions. The line between fantasy and reality in the foundations of physics is the line between the uncomputable and the computable — between breadth and depth/valuation.

8. Conclusion

The 2D continuum model — “uncountability is depth or breadth” — was a useful starting point. But it misses a crucial third axis: **valuation**. The p-adic completions of $\mathbb{Q}$ are neither depth (they are not Archimedean, not connected) nor breadth (they generate falsifiable predictions). They are a distinct category of continuum completions whose physical role is to encode discrete, hierarchical, information-theoretic structure.

The complete picture:

COMPLETIONS OF $\mathbb{Q}$ (Ostrowski)

Archimedean (∞ -place)	2-adic $\mathbb{Q}_2$	3-adic, 5-adic, ... $\mathbb{Q}_3, \mathbb{Q}_5, \dots$
DEPTH (connected, dynamical, causal)	VALUATION (spin, statistics)	VALUATION (color, additional quantum numbers)
BREADTH (non-comp. reals are vacuous)	BREADTH (non-comp. p-adic nums are vacuous)	BREADTH (non-comp. p-adic nums are vacuous)

The physical continuum is not $\mathbb{R}$. It is $\mathbb{R}_c$ together with $\mathbb{Q}_p^c$ at finitely many relevant primes. Depth stays. Breadth goes. Valuation enters — not as geometry, but as information.

The OC program’s ultimate claim: **all physically real entities are generated from finite D/R procedures over $\mathbb{Q}$, completed at finitely many places.** The dark continuum — uncountability-as-breadth — is a mathematical ghost. The connected continuum — uncountability-as-depth — is physically real but countably realized. And the p-adic dimensions — the third axis the 2D model missed — are the missing link between quantum discreteness and number-theoretic structure.

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